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MASTER'S DEGREE IN COMPUTER ENGINEERING

**APEROL from Networks:
Analyzing Pipeline and Embedding Representations
for Optimized Learning (from Networks)**

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1 Introduction

Graphs have emerged as a natural model for the representation and analysis of data and complex systems in various domains. For instance, link prediction in social networks can help to understand the spreading process of rumors or epidemics [1], and in biological networks, it is commonly used for predicting novel interactions between proteins [2].

However, most traditional machine learning algorithms are not designed to work directly with graph-structured data, as they require numeric vectors or matrices as inputs. To address this challenge, embedding techniques have been developed to learn low-dimensional vector representations of nodes, edges, or entire graphs while preserving their topological properties.

In order to design an effective embedding technique, it is necessary to consider several criteria [3]. Two of these are adaptability and scalability. An embedding method is considered adaptable if it can be utilized for various data and tasks. The scalability of an embedding technique is determined by its capacity to process large-scale networks within a reasonable amount of time.

The objective of this study is to evaluate known embedding methods from the perspective of these two criteria.

From the perspective of adaptability, the objective is to determine the potential impact of the graph domain on the precision of an embedding method. In other words, the objective is to ascertain whether there exist approaches that are better suited for a particular field, whether there is one embedding technique that consistently outperforms the others, or if the methods are comparable.

In the context of scalability, the objective is to identify the embedding methods that are better suited to process large graphs. This is accomplished by evaluating the tradeoff between the accuracy of the predictions and the time required for training and inference.

TO DO: structure (se c'è spazio)

2 Embedding methods

Node embedding is the task of mapping nodes into an embedding space, where each node v is represented by a vector $\mathbf{z}_v \in \mathbb{R}^d$ using the mapping function $ENC : v \rightarrow \mathbf{z}_v$. The encoding of nodes into embeddings has to be such that similarity between each pair of nodes is kept in the correspondent embeddings. In order to achieve this, a similarity score $S(u, v)$ is computed between pairwise nodes and a decoder function $DEC : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ is used for computing a similarity score for the embeddings. The discrepancy between the two scores is then compared and minimized by means of a loss function, such as the empirical loss:

$$\mathcal{L} = \sum_{i,j} \ell(DEC(\mathbf{z}_u, \mathbf{z}_v), S(u, v))$$

The node embedding methods are defined by the characteristics of the encoder and decoder.

In this project, four of these methods are considered:

- Node2Vec [4], a shallow encoding method and essentially a tunable version of the DeepWalk algorithm, which is often used in the scientific literature as the baseline for comparing node embedding methods;
- LINE [5], a shallow encoding which is an highly-scalable approximation algorithm;
- Deep Variational Network Embedding in Wasserstein Space (DVNE) [6], a deep encoding method using a Variational Autoencoder. This method has a particular approach of encoding each node to a probability distribution, so in this case the similarity score between embeddings is a distance between probability distributions, which can be measured using the Wasserstein distance;
- GraphSage [7], a deep encoding method making use of a GNN that aggregates the features of each node's neighborhood and learns to generate node embeddings from that.

3 Link Prediction

In this study, the main task considered is link prediction. The link prediction task is defined as the estimation of the existence of a link between two nodes in a network, based on the observed links. Formally,

Network	$ V $	$ E $	Directed	Weighted
Pennsylvania [11, 12]	1,088,092	3,083,796	✓	✗
Padua (province) [13]	122,728	304,184	✓	✓
Hong Kong (city) [14]	43,620	91,542	✓	✓
Italian Covid-19 Retweet Network [15]	221,574	2,332,721	✓	✓
Deezer [16, 17]	54,573	846,915	✗	✗
GitHub Developers [18, 19]	37,700	289,003	✗	✗
Mus Musculus Protein Interactions [20] ¹	20,969	808,988	✗	✓
Saccharomyces cerevisiae Protein Interactions [20] ²	5,786	103,831	✗	✓
Bio-grid-fission-yeast [21]	2,026	12,637	✗	✗

Table 1: Datasets used in the project

given a graph $G = (V, E)$, where V is the set of nodes and E the set of edges, the link prediction task aims to predict whether an edge $(u, v) \in (V \times V) \setminus E$ exists or not, based on the observed edges in E .

In order to perform link prediction using node embeddings, a common approach is to compute a feature vector for each pair of nodes (u, v) based on their embeddings $\mathbf{z}_u$ and $\mathbf{z}_v$. Subsequently, the feature vectors are utilized as input to a binary classifier that predicts the presence or absence of a link between the two nodes.

It is important to note that the quality of the predictions is potentially dependent on two factors: the embeddings themselves and the classifier utilized. To this end, classifiers of varying complexity have been considered:

- Support Vector Machines (SVM) [8], as linear baseline;
- Random Forest (RF) [9], for its really fast computation;
- Multilayer Perceptron (MLP) [10], for accurate predictions.

4 Datasets

The datasets that used in this project are nine and they are divided into three categories: road networks, social networks, and biological networks. For each of these categories, three datasets of increasing size were selected: one small, one medium, and one large.

The datasets, with their main properties (i.e., the number of nodes, the number of edges, if it is directed, and if it is weighted), are presented in Table 1.

5 Experiments

The link prediction task requires a sample of edges from the graph (positive edges) and a sample of edges from the complement graph (negative edges, in this case, do not include self-loops). Consequently, each dataset is partitioned as follows.

Consider graph $G = (V, E)$, where V denotes the set of vertices and E is the set of edges. A first split of the set of edges E is needed for the embedding algorithms, creating E_{embed} and E_{pred} .

E_{embed} is used for the subgraph $G_{\text{embed}} = (V, E_{\text{embed}})$ given as input to the embedding methods: the shallow embedding methods use it to learn a lookup table Z of node embeddings, whereas the deep embedding methods aim to learn a function f that can generalize well.

The remaining edges in E_{pred} are instead reserved for the link-prediction task, where the second split occurs: from E_{pred} the sets E_{train} , E_{val} , E_{test} are created for the training, validation and test of the prediction model. More precisely, E_{train} is used to train the prediction models, E_{val} to select one of them. After that, the selected model is retrained using both E_{train} and E_{val} . Finally, E_{test} is used to evaluate its performance.

¹Confidence score of > 0.4

²Confidence score of > 0.7

In order to achieve a balanced distribution of negative edges, each set contains an equal number of positive and negative examples. Furthermore, in order to avoid data leakage, the sampled negative edges must be distinct from those used internally by the embedding algorithms during training.

The input of the prediction models are edge embeddings. For each edge (u, v) , its corresponding embedded representation is constructed by concatenating the embeddings z_u, z_v of its incident nodes.

The split used in this study is: (from E) 80% for E_{embed} and 20% for E_{pred} ; (from E_{pred}) 60% for E_{train} , 20% for E_{val} and 20% for E_{test} .

5.1 Metrics

Two metrics that are generally used for link prediction are the Area Under the Receiver Operating Characteristic curve (AUROC), and the Area Under the Precision-Recall curve (AUPR). AUROC is historically considered the primary performance metric used to evaluate the performances of link prediction methods [22]. However, AUROC favors accurate classification of positive examples, at the cost of misclassifying the negative ones. In a scenario like the link prediction problem, which is inherently biased towards the negative class, this approach may not be optimal and can overestimate the performance of the methods. Consequently, the AUPR curve was also selected, as it can provide better evaluation of link prediction in the presence of class imbalance.

The values returned by both metrics range from 0 to 1. The AUROC with a value of 1 is indicative of perfect classification, while 0.5 represents random guesses. Similarly, an AUPR value of 1 indicates perfect precision and recall, whereas a value equal to the ratio of positive samples to the total number of samples corresponds to random predictions; in this case, we consider balanced datasets, so the random baseline is 0.5.

For the validation phase, the AUROC metric is used to select the best model, while for the final evaluation both AUROC and AUPR are reported.

5.2 Embedding and classifiers implementation

In regard to the implementation of the embedding techniques mentioned in Sec 2, the following implementations will be adopted: [23] for Node2Vec, [24] for LINE, [25] for DVNE, [26] for GraphSage. Depending on specific needs, some of these could be modified or used as an inspiration for different implementations. In particular, DVNE was not evaluated on directed graphs in the original work [6], meaning it may be necessary to explore adaptations for this setting. If no satisfactory solution is found, DVNE will be restricted only for undirected graphs to maintain a fair comparison.

- Node2Vec [23]
- LINE [24]: the original implementation was optimized for computational efficiency and training stability (for example, using sparse embeddings, SparseAdam optimizer and Batch Matrix Multiplication). In addition, a linear learning rate decay was integrated into the training process. At last, the output generation was modified. In fact, the embeddings for first-order and second-order proximity were trained separately, concatenated into a single representation, and then L2-normalized.
- DVNE [25]: this implementation has been severely rewritten for adapting it to our link prediction pipeline and to make it more similar to the original paper. There are also some changes which resulted in the model being more robust, in particular: the use of log-scaled wasserstein distances, the manual computation of features for the nodes during the training process and using z-score for their standardization. Also a fast approximated method of picking negatives is used for picking the triplets needed in the training process.
- GraphSage [26]: the implementation was modified to support a Graph object as input instead of the original text-based format. In particular, two functions were integrated to generate node features and training labels directly from the graph topology. In the end, the training process was adjusted to return the trained model.

classifiers

- SVM
- MLP: simple model using just 1 layer with 16 hidden channels, cross entropy loss and Adam optimizer.
- RandomForest: for this classifier was used the XGBRFClassifier implementation. This model integrates the Random Forest algorithm within the XGBoost framework, utilizing the histogram-based tree method (hist) to accelerate training on the GPU.

5.3 Experimental setup

The experiments will be conducted using the DEI cluster "Blade". The specifications for the nodes of the cluster are available in the open documentation [27]. More information about the project's memory usage and runtime will be available in the final report.

6 Results

Table 2 reports the AUROC and AUPR values of the best performing classifier for each embedding technique across all datasets.

It can be seen that Node2Vec outperforms the other embedding techniques across the majority of the datasets; however DVNE shows competitive results, surpassing Node2Vec in datasets such as Hong Kong and Italian Covid-19 Retweet Network.

Instead, LINE proves to be the worst performing embedding method in almost all scenarios, probably due to a low number of epochs for the training (necessary to maintain a feasible trade-off between performance and computational time/memory complexity) or its limitation in capturing only local (first and second order) proximity, which may fail to represent the global structure of complex and sparse networks.

GraphSage and DVNE demonstrate a stable performance, each of them showing consistency across all categories of datasets. This suggests that these methods are more robust to variations in graph topology. Two notable extremes are the Italian Covid-19 Retweet Network and Deezer. For the first one, all the embedding methods achieve high values of AUPR and AUROC (with the lowest value of AUROC being 0,9202 for GraphSage), indicating a clear community network where links are easily predictable; while for the second one, all embedding techniques show low values, especially LINE, which reaches an AUPR value of 0,3704 (the lowest value of the entire table), likely due to the high noise or sparse connectivity of the graph.

For what concerns classifiers, Random Forest reaches high performance across most embedding techniques and dataset types. Specifically, it is the best performing model for Node2Vec on road networks and for GraphSage in almost all scenarios. On the other hand, MLP remains highly effective when paired with DVNE and with Node2Vec on social networks.

7 Conclusions

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References

- [1] Romualdo Pastor-Satorras et al. "Epidemic processes in complex networks". In: *Reviews of modern physics* 87.3 (2015), pp. 925–979.
- [2] Björn H Junker and Falk Schreiber. *Analysis of biological networks*. Vol. 2. Wiley Online Library, 2008.
- [3] Anthony Baptista et al. "Zoo guide to network embedding". In: *Journal of Physics: Complexity* 4.4 (2023), p. 042001.
- [4] Aditya Grover and Jure Leskovec. *node2vec: Scalable Feature Learning for Networks*. 2016. arXiv: 1607.00653 [cs.SI]. URL: <https://arxiv.org/abs/1607.00653>.

Dataset	Node2Vec			LINE			DVNE			GraphSage		
	Best Model	AUROC	AUPR	Best Model	AUROC	AUPR	Best Model	AUROC	AUPR	Best Model	AUROC	AUPR
Pennsylvania	RF	0,8369	0,7717	MLP	0,5770	0,5840	MLP	0,7959	0,7710	RF	0,6958	0,6286
Padua	RF	0,8973	0,8551	RF	0,5805	0,5492	RF	0,7761	0,7121	RF	0,8083	0,7343
Hong Kong	RF	0,8278	0,7649	MLP	0,6157	0,6219	MLP	0,8694	0,8464	RF	0,8365	0,7643
Italian Covid-19 Retweet Network	MLP	0,9596	0,9666	MLP	0,9513	0,9575	MLP	0,9716	0,9756	RF	0,9202	0,8808
Deezer	MLP	0,7191	0,5652	RF	0,5415	0,3704	MLP	0,6888	0,5480	RF	0,5769	0,3986
GitHub Developers	MLP	0,9225	0,8865	RF	0,6932	0,5578	MLP	0,9063	0,8540	MLP	0,7239	0,4947
Mus Musculus Protein Interaction	RF	0,8213	0,6877	RF	0,7500	0,6121	RF	0,8091	0,6649	RF	0,7517	0,5762
Saccharomyces cerevisiae Protein Interaction	RF	0,8805	0,8130	RF	0,8005	0,7153	RF	0,8601	0,7491	RF	0,8411	0,7177
Bio-grid-fission-yeast	MLP	0,8754	0,7962	MLP	0,7773	0,6829	RF	0,8636	0,7541	MLP	0,8622	0,7764

Table 2: AUROC and AUPR values for the embedding methods

- [5] Jian Tang et al. “LINE: Large-scale Information Network Embedding”. In: *Proceedings of the 24th International Conference on World Wide Web*. WWW ’15. International World Wide Web Conferences Steering Committee, May 2015, pp. 1067–1077. DOI: 10.1145/2736277.2741093. URL: <http://dx.doi.org/10.1145/2736277.2741093>.
- [6] Dingyuan Zhu et al. “Deep Variational Network Embedding in Wasserstein Space”. In: *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. KDD ’18. London, United Kingdom: Association for Computing Machinery, 2018, pp. 2827–2836. ISBN: 9781450355520. DOI: 10.1145/3219819.3220052. URL: <https://doi.org/10.1145/3219819.3220052>.
- [7] William L. Hamilton, Rex Ying, and Jure Leskovec. *Inductive Representation Learning on Large Graphs*. 2018. arXiv: 1706.02216 [cs.SI]. URL: <https://arxiv.org/abs/1706.02216>.
- [8] Corinna Cortes and Vladimir Vapnik. “Support-vector networks”. In: *Machine learning* 20.3 (1995), pp. 273–297.
- [9] Leo Breiman. “Random forests”. In: *Machine learning* 45.1 (2001), pp. 5–32.
- [10] David E Rumelhart, Geoffrey E Hinton, and Ronald J Williams. “Learning representations by back-propagating errors”. In: *Nature* 323.6088 (1986), pp. 533–536.
- [11] Jure Leskovec et al. “Community structure in large networks: Natural cluster sizes and the absence of large well-defined clusters”. In: *Internet Mathematics* 6.1 (2009), pp. 29–123.
- [12] Jure Leskovec and Andrej Krevl. *SNAP Datasets: Stanford Large Network Dataset Collection*. <https://snap.stanford.edu/data/roadNet-PA.html>.
- [13] Marco Annunziata. *Padua_Network_dataset_2025*. 2025. URL: https://github.com/Remdoox/Padua_Network_dataset_2025.
- [14] yzengal. *RoadNetwork-China-City*. 2021. URL: <https://github.com/yzengal/RoadNetwork-China-City/blob/main/Hongkong.road-d.tar.gz>.
- [15] V. Mesina et al. *Italian Covid-19 Retweet Network (2020-2022) [Data set]*. 13th International Conference on Complex Networks and Their Applications (CNA2024), Istanbul, Turkey. Zenodo. 2024. URL: <https://doi.org/10.5281/zenodo.13909011>.
- [16] Benedek Rozemberczki et al. “GEMSEC: Graph Embedding with Self Clustering”. In: *Proceedings of the 2019 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining 2019*. ACM. 2019, pp. 65–72.
- [17] Jure Leskovec and Andrej Krevl. *SNAP Datasets: Stanford Large Network Dataset Collection*. <https://snap.stanford.edu/data/gemsec-Deezer.html>.
- [18] Benedek Rozemberczki, Carl Allen, and Rik Sarkar. *Multi-scale Attributed Node Embedding*. 2019. arXiv: 1909.13021 [cs.LG].
- [19] Jure Leskovec and Andrej Krevl. *SNAP Datasets: Stanford Large Network Dataset Collection*. <http://snap.stanford.edu/data/github-social.html>.
- [20] Damian Szklarczyk et al. “The STRING database in 2023: protein–protein association networks and functional enrichment analyses for any sequenced genome of interest”. In: *Nucleic Acids Research* 51.D1 (2023), pp. D638–D646. DOI: 10.1093/nar/gkac1000. URL: <https://string-db.org/cgi/download>.
- [21] Ryan A. Rossi and Nesreen K. Ahmed. “The Network Data Repository with Interactive Graph Analytics and Visualization”. In: *AAAI*. 2015. URL: <https://networkrepository.com>.
- [22] I. Bhargavi Kalyani, A. Rama Prasad Mathi, and Niladri Sett. “Evaluating link prediction: new perspectives and recommendations”. In: *International Journal of Data Science and Analytics* 20.7 (2025), pp. 6855–6886. ISSN: 2364-4168. DOI: 10.1007/s41060-025-00858-0. URL: <https://doi.org/10.1007/s41060-025-00858-0>.
- [23] palash1992. 2022. URL: <https://github.com/eliorc/node2vec>.
- [24] dmpierre. 2019. URL: <https://github.com/dmpierre/LINE>.
- [25] Lakshya-99. 2020. URL: https://github.com/Lakshya-99/Deep_Variational_Network_Embedding/tree/master.
- [26] William L. Hamilton et al. 2018. URL: <https://github.com/williamleif/graphsage-simple/>.

- [27] Università degli Studi di Padova. *Cluster "Blade"*. URL: <https://docs.dei.unipd.it/en/CLUSTER/Overview>.

Contribution of Authors and AI Usage

The contributions for this project are:

- ...

The following AI tools were used:

- Copilot was used to suggest code and report snippets.
- DeepL was used to correct the grammar and syntax of the text.